Course Syllabus

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Georgia Tech
Scheller College
of Business

Instructor	Email	Office Hours
John McSwain, MSHCI, MBA, MSE (https://www.scheller.gatech.edu/directory/faculty/mcswain/index.html)	john.mcswain@scheller.gatech.edu (mailto:john.mcswain@scheller.gatech.edu)	Appointments by request via Calendly.com (https://Calendly.com)

DOWNLOAD PDF VERSION OF SYLLABUS (https://canvas.instructure.com/courses/7100862/files/217604634?wrap=1) (https://canvas.instructure.com/courses/7100862/files/217604634/download?download_frd=1)

Course Information

The MGT 7316 course will teach you how to design, develop, and deploy cloud server(less) applications for consumer and enterprise clients, covering the client-server model, requirements definition, technical architecture, coding, and cloud deployment.

Description

This course will cover the design, development, and deployment of cloud server(less) applications that can serve and support consumer and enterprise client applications. You will learn about the client-server model that governs web communication over HTTP. The first module introduces you to processes and methodologies of defining concise and actionable functional and non-functional requirements. The second module will introduce technical architecture and documentation methods. The third module will guide the development of a coded/programmed representation of the solution. The fourth module will provide you with opportunities to deploy their solutions to the cloud. The fifth module will allow you to collaboratively produce a business problem solution within a working group.

Prerequisites

Intermediate or higher proficiency with at least one of the following programming languages: JavaScript (client-side or server-side Node.js), Python 3+, Go, or Java—familiarity with cloud provider technology, specifically Google Cloud Platform.

Learning Objectives and Assignments

By completing all course requirements, you will be able to:	List and brief explanation of activities validating outcome achievement for each objective:
Define and compose business problem statements	Module 1 Assignment: Product Requirements Document (PRD) (https://canvas.instructure.com/courses/7100862/assignments/38364328) The activity will introduce you to requirement definition, scope management, decision-making, and documentation standards.
Enact actionable methodologies for business operational/tactical/strategic decision-making and problem solving	Module 2 Assignment: Web Application Solution Architecture & Application Design Document (SADD) (https://canvas.instructure.com/courses/7100862/assignments/38364402) The activity will introduce you to architectural descriptions, design principles, (reusable) component and user interface design, data modeling, error handling, and unit testing.
Utilize programming tools, design methodologies, and syntax that enacts computer (en)coded business solutions	Module 3 Assignment: Web Application Development (https://canvas.instructure.com/courses/7100862/assignments/38364518) Module 4 Assignment: Web Application Solution Testing and Deployment (https://canvas.instructure.com/courses/7100862/assignments/38364521) The activities will introduce you to performance, security considerations, development patterns, and processes. Finally, the activity will guide you to experience deployment, infrastructure reflection, and ongoing development and maintenance.

Course Materials

Required Text

[Docs for Devs (available via GT Online Library)](https://tinyurl.com/docsfordevs) (https://tinyurl.com/docsfordevs)
[Mozilla Developer Network JavaScript Guide](https://developer.mozilla.org/en-US/docs/Learn/JavaScript) (https://developer.mozilla.org/en-US/docs/Learn/JavaScript)
[Google Technical Writing](https://developers.google.com/tech-writing) (https://developers.google.com/tech-writing)
[Google for Developers](https://developers.google.com/) (https://developers.google.com/)
[Google Cloud Computing](https://cloud.google.com/) (https://cloud.google.com/)

Other Readings

[JavaScript Data Structures, Client-side Frameworks](https://developer.mozilla.org/en-US/docs/Web/JavaScript/Data_structures%20https://developer.mozilla.org/en-US/docs/Learn/Tools_and_testing/Client-%20side_JavaScript_frameworks) (https://developer.mozilla.org/en-US/docs/Web/JavaScript/Data_structures%20https://developer.mozilla.org/en-US/docs/Learn/Tools_and_testing/Client-%20side_JavaScript_frameworks)
[W3Schools – JavaScript Reference](https://www.w3schools.com/js/default.asp) (https://www.w3schools.com/js/default.asp)

Technical Requirements

- Reliable and available on-demand Internet-Access
  - Upload/download speeds of 10+ Mbps or higher
- Computer System Requirements
  - 8+ GB of RAM, ~20 GB of free/available hard drive space Windows - Windows 10, 11+ (64-bit)
  - Mac - macOS Big Sur (11) or higher
  - Linux - 64-bit kernel (Ubuntu 16.04+, Debian 9+, Fedora 37+)

Software Applications & Accounts

- Install [Zoom Desktop Client](https://zoom.us/) (https://zoom.us/)
- (https://zoom.us/) Download [Diagrams.net Desktop Application](https://get.diagrams.net/) (https://get.diagrams.net/) or [Diagrams.net Browser App](https://get.diagrams.net/) (https://get.diagrams.net/)
- Create a free (or paid) [Replit](https://replit.com/) (https://replit.com/) account
- Create a free (or paid) [Google](https://accounts.google.com/signup) (https://accounts.google.com/signup) account

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Learning Environment & Activities

This course will be delivered online through the LMS course management system. Requirement documentation, developer documentation, and source code development occur within the Replit online integrated development environment. Modules will span two calendar weeks per topic. See the [Course Rubrics](https://canvas.instructure.com/courses/7100862/rubrics) (https://canvas.instructure.com/courses/7100862/rubrics) for submission quality guidelines.

Participation: Discussions

Homework assignments and discussion posts are assigned and due throughout the semester. Assignments are combinations of programming, reading, and documentation. You will submit URLs referencing online repositories that store code and documentation deliverables and, in some cases, may require nested links to live/online services. You are responsible for submitting properly formatted links and URLs. Incorrectly formatted links or unreadable/corrupted files will receive no credit. All homework submissions should adhere to the Documentation and Code Rubric.

While there is no formal attendance policy, the course requires discussion posts of (1) initial post thread and at least (2) replies per week (3+ posts x 2 weeks = 6+ posts per module) and should adhere to the discussion rules below:

Discussion Quality Rubric (For full credit consideration)

- **Quality**
  Appropriate comments: thoughtful, reflective, and respectful of other’s postings
- **Relevance**
  Posting thoroughly answers the discussion prompts and demonstrates understanding of material with well-developed ideas and integrates assigned content and makes strong connections to practice
- **Mechanics**
  Writing is free of grammatical, spelling, or punctuation errors.

Pre- and Post- Assessment quizzes

During the first week of the course, a pre-assessment quiz will identify any pre-existing knowledge which can be used as a baseline to adjust the course trajectory and add any personalized content or objectives, if needed. During the last week of the source, a post-assessment quiz will measure any variance between desired and actual learning outcomes.

Homework

Bi-weekly homework assignments are assigned and are due throughout the semester. Assignments may be a combination of programming, reading, and written papers regarding topics covered in class. You will submit URLs BEFORE the due date (e.g., 11:59 pm ET). You are responsible for submitting properly formatted files (e.g., written documents, code URLs). Incorrectly formatted links or unreadable/corrupted files will receive no credit.

Team Project

Project teams consist of student-formed teams of 3-5 people and provide an opportunity to ideate, concept, develop, and deploy a web application that solves a business. The team project combines programming concepts/skills and business modeling topics covered throughout the semester. Project requirements (e.g., checkpoints and deliverables) are due before the final date. Collaboration is allowed within the team, not across teams. Peer evaluations are required following group project presentations, and individual grades are adjusted accordingly.

Grading

Assignment Descriptions and Weights

Assignment/Activity Name	Description	Points
Pre-assessment Quiz (https://canvas.instructure.com/courses/7100862/quizzes/14680142)	Learner baselining	10
Discussion Posts (https://canvas.instructure.com/courses/7100862/discussion_topics)	Includes module-specific discussion threads and 30 replies, totaling 3+ per week, 6+ per module	30
Module 1 Assignment (https://canvas.instructure.com/courses/7100862/assignments/38364328)	Generate a web application product requirements document (PRD) using **Markdown (https://www.markdownguide.org/getting-started/)** syntax	30
Module 2 Assignment (https://canvas.instructure.com/courses/7100862/assignments/38364402)	Generate an solution architecture and application design document, based on requirements generated in Module 1, using **Markdown (https://www.markdownguide.org/getting-started/)** syntax	30
Module 3 Assignment (https://canvas.instructure.com/courses/7100862/assignments/38364518)	Write application code to implement the application design generated from Module 2.	30
Module 4 Assignment (https://canvas.instructure.com/courses/7100862/assignments/38364521)	Deploy web application from Module 3 to the Google Cloud Platform	30
Module 5 Assignment (https://canvas.instructure.com/courses/7100862/assignments/38364537)	Work with an assigned group to compose a PRD and SADD, implement/code, and deploy a functional web application	60
Post-assessment Quiz (https://canvas.instructure.com/courses/7100862/quizzes/14682562)	Measure objective differential	10

Your final grade in this course is a letter grade based on your total percentage grade.

Grading Scale

Letter Grade	Percentage/Points
A	>= 90.0% (>= 207 pts)
B	>= 80% & < 90% (>= 184 pts & < 207 pts)
C	>= 70% & < 80% (>= 161 pts & < 184 pts)
D	>= 60% & < 70% (>= 138 pts & < 161 pts)
F	< 60% (< 138 pts)

Scores will NOT be rounded up to achieve a higher, preferable letter grade. Points tallied will be scored YOU have EARNED.

Assignment Submission Procedures

All submissions will be made using public Replit URLs to the LMS. Discussion posts will be made directly to the LMS.

Grade Discussion Policy

You may request individual assignment score reviews regarding assignments within ONE CALENDAR WEEK from the day grade posts on the LMS. If requested, please be prepared to discuss quality concerning the grading rubric. Once the grade discussion week window has closed, the grade for an assignment is FINAL.

Late Submission & Missed Assignments

No late work will be accepted, and no make-up work will be provided. Due dates are posted and enforced in Canvas. If preconditions could lead to late submissions, send an email or documentation, including Georgia Tech-sponsored events BEFORE the assignment's due date. Otherwise, requesting late submission consideration will be referred to **Georgia Tech's Office of Student Life (https://studentlife.gatech.edu/)**, where documentation is required for assignment reconsideration.

Course Policies

Academic Integrity

This course will follow the guidelines established by **Georgia Tech's Academic Honor Code (http://www.catalog.gatech.edu/policies/honor-code/)** and **Code of Conduct (https://catalog.gatech.edu/rules/18/)**. All sources of the information quoted in any course assignments must be appropriately acknowledged. Please keep in mind that academic dishonesty includes (a) cheating, (b) fabrication and falsifications, (c) multiple submissions, (d) plagiarism, and (e) complicity in academic dishonesty. Violations of the code in any form will be addressed to the full extent allowed by the Honor Code. You will be asked to acknowledge their acceptance of this stipulation and willingness to abide by all terms of the Honor. For information on Georgia Tech's Academic Honor Code, please visit **http://www.catalog.gatech.edu/policies/honor-code/ (http://www.catalog.gatech.edu/policies/honor-code/)** or **http://www.catalog.gatech.edu/rules/18/ (http://www.catalog.gatech.edu/rules/18/)**.

Communication and Official Correspondence

All course-related materials, including announcements, grades, lecture notes, assignments, course schedule updates, and additional teaching materials, will be posted on the class website on Canvas. It is the your responsibility to check the class website regularly (especially the evenings before class). Canvas announcements and email will be the methods of official correspondence for this course. Emails will be sent to your official email addresses registered with Georgia Tech (which usually end in @gatech.edu). Each student's responsible for having his/her Georgia Tech email account active and checking emails sent to that address regularly (preferably at least once a day). If you prefer to use a personal email account (like Gmail or Yahoo), forward your Georgia Tech email to that address and ensure it is not filtered as spam.

Student-Faculty Agreement

At Georgia Tech, striving for an atmosphere of mutual respect, acknowledgment, and responsibility between faculty members and you is essential. See the GT catalog for an articulation of some basic expectations between instructor and you. Ultimately, respect for knowledge, hard work, and cordial interactions will help build our desired environment. You are encouraged to remain committed to the ideals of Georgia Tech while in this class.

Subject to Change Statement

The syllabus and course schedule may be subject to change. Changes to the course, dates, or content are communicated via LMS announcements. It is your responsibility to stay current.

Student Feedback

During this course, I may ask for feedback about your learning in informal and formal ways. I value your thoughts on what we do in class, so I encourage you to respond to feedback requests to help co-create an effective learning environment.

Special Needs & Accommodations for Students with Disabilities

If you are a student with learning needs that require special accommodation, contact the Office of Disability Services at (404)894-2563 or **http://disabilityservices.gatech.edu/** (http://disabilityservices.gatech.edu/) as soon as possible to make an appointment to discuss your needs and to obtain an accommodations letter. Please also email me as soon as possible to set up a time to discuss your learning needs.

Tentative Course Schedule

Course Schedule

Module	Topics/Readings	Activities	Assignments/Due Dates
1	**Requirements Engineering** (https://canvas.instructure.com/courses/7100862/modules/12817824)	**Web Tools** Replit integrated development environment **Lectures** Course Introduction Slides Wicked Business Problems Slides Defining “users”, “requirements”, and “done” Slides **Readings** **Product Requirements Docs (Atlassian)** (https://www.atlassian.com/agile/product-management/requirements) **Google Technical Writing One** (https://developers.google.com/tech-writing/one/documents) **Markdown Guide** (https://www.markdownguide.org/) **Assignments** **Pre-assessment Quiz** (https://canvas.instructure.com/courses/7100862/quizzes/14680142) **Module 1 Discussion Post** (https://canvas.instructure.com/courses/7100862/discussion_topics/18617763) **Product Requirement Document (PRD)** (https://canvas.instructure.com/courses/7100862/assignments/38364328)	Weeks 1 through 2 Assignment: Due by end of Week 2
2	**Web Application Solution Architecture and Design** (https://canvas.instructure.com/courses/7100862/modules/12817828)	**Web Tools** Replit integrated development environment **Lectures** Model-View-Controller Slides Client-Server Model Slides Microservice Pattern Slides **Readings** Docs for Devs (Chapter 6) **Software Architecture Guide** (https://martinfowler.com/architecture/) **Model-View-Controller (Mozilla Developer Network)** (https://developer.mozilla.org/en-US/docs/Glossary/MVC) **Assignments** **Module 2 Discussion Post** (https://canvas.instructure.com/courses/7100862/discussion_topics/18617878) **Web Application Solution Architecture and Design Document (SADD)** (https://canvas.instructure.com/courses/7100862/assignments/38364402)	Weeks 3 through 4 Assignment: Due by end of Week 4
3	**Web Application Programming and Development** (https://canvas.instructure.com/courses/7100862/modules/12817832)	**Web Tools** Replit integrated development environment	Weeks 5 through 6 Assignment: Due by end of Week 6

		Lectures Web Programming Pattern Slides Serverless Slides Module Pattern Slides **Readings** OOP (MDN) (https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide) JavaScript Patterns (https://shichuan.github.io/javascript-patterns/) Google Cloud Functions (https://cloud.google.com/functions) Google Cloud Storage: Firebase (https://firebase.google.com/docs/storage) Google Cloud Storage: Buckets (https://cloud.google.com/storage/docs/creating-buckets) **Assignments** Module 3 Discussion Post (https://canvas.instructure.com/courses/7100862/discussion_topics/18617913) Create/Submit Web Application source code (https://canvas.instructure.com/courses/7100862/assignments/38364518)	
4	Web Application Solution Testing Deployment (https://canvas.instructure.com/courses/7100862/modules/12817834)	**Web Tools** Replit integrated development environment **Lectures** Cloud Deployment Pattern Slides Unit/Functional Testing Slides Refactoring Slides **Readings** Practical Test Pyramid (https://martinfowler.com/articles/practical-test-pyramid.html) Google Cloud Deploy (https://cloud.google.com/deploy) Refactoring (https://galileo-gatech.primo.exlibrisgroup.com/permalink/01GALI_GIT/jb7gbs/cdi_safari_books_v2_9780134757681) **Assignment** Module 4 Discussion Post (https://canvas.instructure.com/courses/7100862/discussion_topics/18617930) Create/Submit Web Application testing source code and deployment pipeline (https://canvas.instructure.com/courses/7100862/assignments/38364521)	Weeks 7 through 8 Assignment: Due by end of Week 8
5	Team Project (https://canvas.instructure.com/courses/7100862/modules/12817837)	**Web Tools** Replit integrated development environment **Lectures** Agile / Scrum / Kanban Slides **Readings** Agile Project Management (https://www.atlassian.com/agile/project-management) Kanban (https://www.atlassian.com/agile/kanban) Scrum (https://www.atlassian.com/agile/scrum) **Assignment** Post-assessment Quiz (https://canvas.instructure.com/courses/7100862/quizzes/14682562) Module 5 Discussion Post (https://canvas.instructure.com/courses/7100862/discussion_topics/18617932) Project Submission (PRD, SADD, Source code, Deployed URL) (https://canvas.instructure.com/courses/7100862/assignments/38364537)	Weeks 9 through 10 Assignment: Due by end of Week 10

Course Summary:

Date	Details	Due
	Module 5: Team Project (https://canvas.instructure.com/courses/7100862/assignments/38364537)	
	Module 1 Assignment: Product Requirement Document (PRD) (https://canvas.instructure.com/courses/7100862/assignments/38364328)	
	Module 2 Assignment: Solution Architecture and Design Document (SADD) (https://canvas.instructure.com/courses/7100862/assignments/38364402)	
	Module 3: Web Application Programming and Development (https://canvas.instructure.com/courses/7100862/assignments/38364518)	
	Module 4: Web Application Solution Testing and Deployment (https://canvas.instructure.com/courses/7100862/assignments/38364521)	
	Post-assessment Quiz (https://canvas.instructure.com/courses/7100862/assignments/38412913)	

Date	Details	Due
	Pre-assessment Quiz (https://canvas.instructure.com/courses/7100862/assignments/38407897)	